Emerging Paradigms in Quantitative Finance: Integrating Binomial Pricing, Long Short-Term Memory, Reinforcement Learning, and Regime Detection within the Framework of Implied Volatility

Palash Shah, Carnegie Mellon University

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1 Introduction

You are reading this sentence without knowing exactly how it will end, yet you have a remarkably sharp sense of what it is unlikely to do. It will probably not dissolve into a recipe, switch to Mandarin, or erupt into pure noise. You cannot predict the next word, but you can anticipate the rough corridor within which the sentence is likely to wander—its style, register, and tone. When you first opened this paper, you likely did something similar: glanced at the title, skimmed the table of contents, or checked my affiliations before deciding whether it was worth your time. That behavior reflects the same distinction that underlies volatility: not forecasting precise outcomes, but constraining the distribution from which they are drawn.

In financial markets, price paths exhibit the same logic. Market participants talk in terms of bullish or bearish narratives, but what is traded, hedged, and systematically engineered is the variability around those narratives. Direction is a story; volatility is the budget for how wrong that story can be. Options markets make this explicit: every implied volatility quote operationalizes a view not just about where prices might end, but about how violently they may traverse the space of possibilities along the way.

This paper adopts the stance that, in options markets, one does not merely trade direction with volatility as an afterthought—one trades volatility as the primary object. The implied-volatility surface is treated as a high-dimensional state variable encoding information about level, skew, term structure, and tail behavior, rather than as a static input to a pricing formula. The central questions become: which volatility regime currently prevails, what does the surface imply about asymmetry and extreme events, and how should capital be scaled and allocated conditional on that regime?

Methodologically, my work proposes a synthesis of classical mathematical finance with modern machine learning. Stochastic volatility models and risk-neutral valuation provide the structural backbone for interpreting option prices and their deformations. On this foundation, spectral clustering is applied to implied-volatility configurations to identify latent regimes on the volatility manifold, long short-term memory networks are trained to forecast the evolution of this manifold, and reinforcement learning agents map these forecasts into trading policies that adapt position size and structure to regime-specific risk–return profiles.

The objective is to develop a unified algorithmic framework in which the implied-volatility surface defines the state space, regimes emerge from its geometry, and trading policies exploit statistically significant, regime-consistent mispricings in volatility.

2 The Central Role of Implied Volatility in Modern Markets

2.1 Background: Why Volatility Dominates

Implied volatility arises naturally as a free variable within option models. In liquid option markets, participants do not directly price volatility; rather, they bid and ask on options contracts, and

the volatility that equilibrates theoretical price and market price for an option is inferred as the "implied volatility." This scalar serves as risk barometer, consensus forecast, and medium of tactical speculation. Its centrality is established both by mathematical derivation and by its empirical omnipresence—as seen in the CBOE Volatility Index.

2.2 Mathematical Definition

By definition, let C_{mkt} be the observed price of a European call, and $C_{\text{model}}(S, K, T, r, \sigma)$ its theoretical price under a parametric model. Then the IV (σ_{imp}) is defined by the root of the equation:

$$C_{\text{mkt}} = C_{\text{model}}(S, K, T, r, \sigma_{\text{imp}}). \tag{1}$$

This implicit equation may admit more than one solution for illiquid or mispriced options, but under mild assumptions uniqueness holds.

2.3 Why IV is Fundamental

The fundamental theorem of asset pricing, in its mathematical form, guarantees the existence of a risk-neutral probability measure under which discounted prices of tradeable securities are martingales. The centrality of IV is its role as the "plug-in" statistic translating option market consensus into forecastable future uncertainty, marking it as the unique free risk parameter in a risk-neutral pricing world.

3 The Mathematical Foundations of Implied Volatility

Implied volatility is more than just an input or calibration parameter; it is, as will be shown, the key mathematical unknown in almost every model of option price, risk, and premium. To fix ideas, consider the Black-Scholes model:

$$C = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2) \tag{2}$$

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{3}$$

$$d_2 = d_1 - \sigma\sqrt{T}, \tag{4}$$

where S_0 is spot price, K strike, T time to expiration, r risk-free rate, and σ volatility. All variables but σ are either observable or contractually set. The non-linearity of $C(\sigma)$ ensures that for any C_{mkt}, there is generically a unique σ_{imp} such that $C_{\text{mkt}} = C(\sigma_{\text{imp}})$.

Implied volatility thus inverts the option pricing machinery: Instead of using volatility to price options, the market price is used to infer volatility.

3.1 Properties and Theorems

Proposition: (Uniqueness of Implied Volatility, under monotonicity)
Let $C(\sigma)$ be the Black-Scholes pricing function for a European call option, strictly increasing in σ for positive σ and all other parameters fixed. Then, for each C_{mkt} in the image of $C(\cdot)$, there exists a unique implied volatility σ_{imp}.

Proof: Note that $\frac{\partial C}{\partial \sigma} > 0$ for $\sigma > 0$, per Vega in Black-Scholes. Thus, $C(\cdot)$ is injective on $(0, \infty)$. The intermediate value theorem applies. $\square$

3.2 Smile, Surface, and Market Impact

Empirically, IV is not a constant, even for a single underlying: Different strikes and maturities trade at different implied volatilities, forming the volatility smile and volatility surface. This is observed directly in index options such as the S&P 500, whose options market generates a smooth but undulating map of IV as a function of strike and maturity.

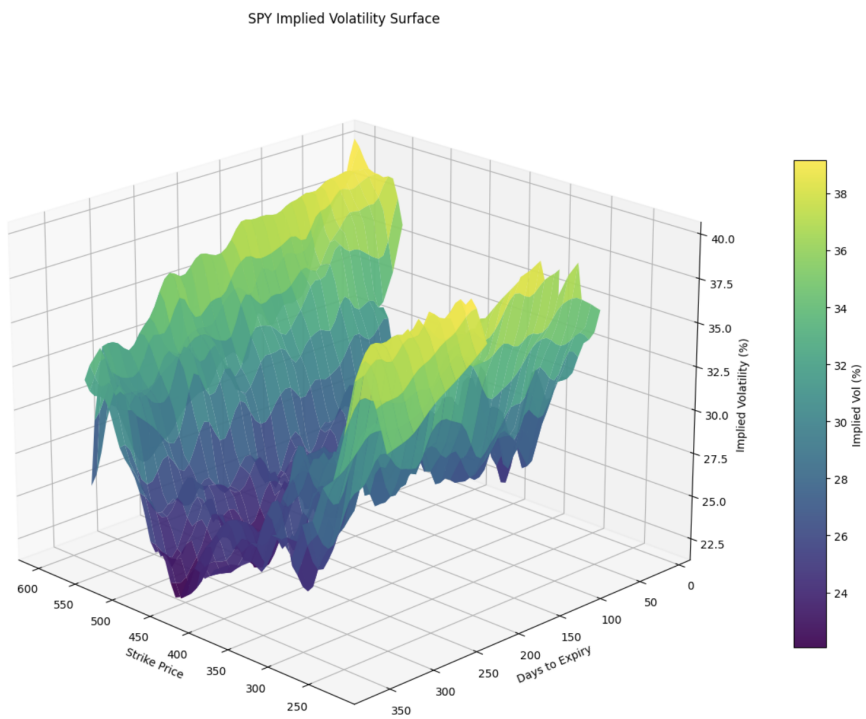


Figure 1: The surface highlights typical market skew dynamics as well as term structure effects observed in liquid equity index options. For full code and data, see this Kaggle notebook.

4 Formal Option Pricing Models and Their Dependence on Implied Volatility

4.1 Cox-Ross-Rubinstein Binomial Model

The binomial model, independently formalized by Cox, Ross, and Rubinstein, provides a discrete-time, recombining tree construction for modeling the evolution of the underlying security. The model postulates that, in each of N time steps, the asset price S may move up by a multiplier u or down by d:

$$u = e^{\sigma\sqrt{\Delta t}} \tag{5}$$

$$d = e^{-\sigma\sqrt{\Delta t}} \tag{6}$$

$$p = \frac{e^{r\Delta t} - d}{u - d} \tag{7}$$

where p is the risk-neutral probability.

To price a European option, follow the tree forward, then discount the expected payoff at maturity backward through the tree by risk-neutral probabilities. The link to IV becomes clear upon calibration: one adjusts σ until the modeled price matches market observation.

4.2 Relationship to Black-Scholes

It is proven that as $N \to \infty$, the binomial model's price converges to the price generated by the Black-Scholes formula, assuming all else equal, thus inheriting the IV inversion properties of Black-Scholes in the continuous limit.

4.3 American Option Pricing in the Binomial Model

American options, exercisable before expiry, are naturally handled in the binomial model. At each node, the price is:

$$V_i^n = \max\left\{\text{intrinsic value at node, } e^{-r\Delta t}\left(pV_i^{n+1} + (1-p)V_{i+1}^{n+1}\right)\right\} \tag{8}$$

The critical dependence on the chosen volatility σ remains.

5 Stochastic Processes Underlying Price and Volatility

Modern quantitative finance relies fundamentally on stochastic processes to model both the price evolution of assets and the dynamics of volatility itself. At the root of these models is Brownian motion, formalized as a continuous-time limit of random walks.

5.1 Geometric Brownian Motion (GBM)

Let W_t denote a standard Brownian motion. The dynamics of a stock price under GBM are modeled as:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{9}$$

where μ is drift, σ volatility, and S_t price at time t. The solution to this stochastic differential equation is:

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right] \tag{10}$$

It is the variance parameter σ which underpins both realized and implied volatility.

5.2 Stochastic Volatility Models

In reality, volatility itself fluctuates over time rather than remaining constant. Stochastic volatility models, such as the Heston and SABR families, introduce an additional stochastic differential equation (SDE) for the variance process and explicitly allow the shocks to price and volatility to be correlated. A prototypical specification is

$$dS_t = \mu S_t\, dt + \sqrt{v_t}\, S_t\, dW_t, \tag{11}$$

$$dv_t = \kappa(\theta - v_t)\, dt + \xi\sqrt{v_t}\, dZ_t, \tag{12}$$

with $\mathrm{corr}(dW_t, dZ_t) = \rho$. Here v_t is the instantaneous variance, κ the mean-reversion speed toward long-run level θ, ξ the volatility of variance (vol-of-vol), and ρ the contemporaneous correlation between price and volatility shocks.

The correlation parameter ρ is crucial for generating realistic implied-volatility smiles and skews. When $\rho = 0$, the implied volatility surface produced by such models is close to flat near the money, especially for short maturities; most of the curvature is then driven by higher-order effects rather than a systematic skew. In contrast, a negative ρ (the leverage effect) makes volatility tend to rise when prices fall, which increases the value of out-of-the-money puts and produces the pronounced negative skew observed in equity index options. Parameters like $(\kappa, \theta, \xi, \rho)$ therefore control the dynamic shape of the implied-volatility surface: κ and θ govern level and mean reversion, ξ controls the roughness of volatility, and ρ shapes the slope and curvature of the skew.

This perspective connects directly to the machine learning techniques developed later in the paper. Models such as LSTMs and spectral-clustering-based regime detection can be viewed as nonparametric tools that learn how effective parameters of a latent stochastic-volatility mechanism evolve over time. In particular, the implied-volatility surfaces and their spectral regimes implicitly encode time-varying analogues of $\kappa, \theta, \xi, \rho$, while the LSTM forecasts and reinforcement-learning

policy effectively learn how to exploit predictable structure in the evolution of skew and level without committing to a specific closed-form SDE.

5.3 Theoretical Properties

In stochastic volatility settings, option pricing is still carried out under a risk-neutral measure $\mathbb{Q}$ for which the discounted asset price $e^{-rt}S_t$ is a martingale. Constructing $\mathbb{Q}$ typically requires a Girsanov change of measure from the physical dynamics to risk-neutral dynamics, after which parameters $(\kappa, \theta, \xi, \rho)$ are calibrated so that model-implied option prices match observed market prices. This preserves the fundamental martingale structure of arbitrage-free pricing while allowing for richer volatility behavior than in the Black–Scholes model.

6 Long Short-Term Memory Neural Networks in Volatility Forecasting

6.1 Deep Learning and Volatility

Classic volatility models (GARCH, EWMA) capture only part of the time-series structure found in financial data. LSTM networks, a subclass of recurrent neural networks (RNNs), can learn complex temporal dependencies, including non-linear autocorrelations, long-term memory effects, and volatility regime switches.

6.2 LSTM Cell Dynamics

Let x_t be the input vector at time t, typically including log-returns, historical realized volatility, volume metrics, and technical features. The LSTM cell processes as follows:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \qquad \text{(forget gate)} \tag{13}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \qquad \text{(input gate)} \tag{14}$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \tag{15}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{16}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \qquad \text{(output gate)} \tag{17}$$

$$h_t = o_t \odot \tanh(C_t) \tag{18}$$

Here, σ is the logistic sigmoid, tanh is hyperbolic tangent, W_* and b_* denote weights and biases, and $\odot$ is component-wise multiplication.

6.3 Theoretical Power: Universal Approximation

Hochreiter and Schmidhuber proved (1997) that LSTMs have the power to approximate arbitrary sequence-to-sequence mappings given sufficient data and depth. When applied to financial time series, an LSTM can model the joint distribution of returns, lagged volatility, and prospective realized volatility.

6.4 Volatility Forecasting Process

Typically, the network is trained to predict $y_t = \sigma_{\text{realized},t+\Delta}$, the realized volatility for a lookahead window Δ, using historical rolling windows as inputs. The loss function can be mean squared error, or in advanced settings, probabilistic losses that capture full conditional distributions.

6.5 Mathematical Justification

Consider the process $(x_{t-n}, \ldots, x_t)$ as a Markovian or non-Markovian sufficient statistic for predicting future volatility. The conditional expectation $\mathbb{E}[\sigma_{t+\Delta}|x_{t-n}, \ldots, x_t]$ can be approximated by a sufficiently expressive LSTM, due to universal approximation properties for RNNs.

7 Reinforcement Learning for Adaptive Trading Decisions

7.1 Foundations of Reinforcement Learning in Markets

Reinforcement learning (RL) formalizes sequential decision-making in uncertain, stochastic environments. In trading, an RL agent observes market states, selects actions (enter, hold, exit, adjust position size), and receives reward (P&L, risk-adjusted return).

7.2 Markov Decision Processes

A Markov Decision Process is defined by states S, actions A, rewards R, and transition probabilities P. The objective is to learn a policy $\pi : S \to A$ maximizing expected cumulative reward. The RL problem is typically set up as:

$$Q_\pi(s, a) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = s, a_t = a \right] \tag{19}$$

where Q_π is the value function, γ discount factor, and r_t the reward at step t.

7.3 Optimal Stopping in Trading

For trade exit, the optimal stopping problem can be written as:

$$V_t = \sup_{\tau \geq t} \mathbb{E}\left[e^{-r(\tau - t)} f(S_\tau) \mid \mathcal{F}_t\right] \tag{20}$$

where τ is the stopping time, f is the reward on exit, and $\mathcal{F}_t$ is the filtration of current information.

RL agents approximate the solution through empirical learning of Q-values. In Deep RL, function approximators (neural networks) replace lookup Q-tables for high-dimensional or continuous state spaces.

7.4 Convergence Theorems

Under sufficient exploration and appropriate learning rates, Q-learning converges to the fixed point:

$$Q^*(s, a) = \mathbb{E}\left[r_{t+1} + \gamma \max_{a'} Q^*(s_{t+1}, a') \mid s_t = s, a_t = a\right] \tag{21}$$

almost surely, as shown in Watkins and Dayan (1992).

7.5 Application to Volatility-Driven Decision-Making

While RL does not estimate volatility directly, it adapts to current volatility indirectly by optimizing reward in changing environments. In volatility-adaptive trading, states include realized/forecast volatility, recent returns, and regime tags; actions are entry, hold, exit, and risk adjustment; rewards are P&L, possibly penalized by drawdowns or excess volatility.

8 Market Regime Detection: Concepts and Algorithms

Markets alternate between trend, mean-reversion, high-volatility, and quiet modes. Identifying regimes is crucial for volatility-sensitive strategies.

8.1 Hidden Markov Models

Hidden Markov Models (HMMs) segment time series into hidden states characterized by transition matrices and emission distributions. Let $\{r_t\}$ be returns, $z_t \in \{1, \ldots, K\}$ hidden regime. The probability of regime sequence is:

$$P(z_{1:T}) = \prod_{t=2}^{T} P(z_t | z_{t-1}) \prod_{t=1}^{T} P(r_t | z_t) \tag{22}$$

Baum-Welch algorithm is used for parameter estimation.

8.2 Ensemble Classification

More recently, ensemble ML classifiers (Random Forest, XGBoost, neural nets) take engineered features—rolling returns, volatility, momentum indicators, etc.—to assign regime labels. Each classifier contributes probability, ensemble votes decide regime.

8.3 Statistical Properties

Regimes often show autocorrelation and persistence (self-transition probabilities ¿ 0.7 in typical HMM fits). The regime label is a crucial input for both RL and LSTM submodules, as transitions frequently align with spikes in realized volatility.

9 Advanced Technical Indicators: DEMA and Supertrend

9.1 Double Exponential Moving Average (DEMA)

Standard exponential moving average (EMA) suffers from lag; DEMA improves responsiveness. Let x_t be price at t:

$$\mathrm{EMA}_t = \alpha x_t + (1-\alpha)\mathrm{EMA}_{t-1}, \quad \alpha = \frac{2}{N+1} \tag{23}$$

$$\mathrm{DEMA}_t = 2 \cdot \mathrm{EMA}_t - \mathrm{EMA}(\mathrm{EMA}_t) \tag{24}$$

DEMA reduces lag while maintaining smoothness, yielding faster reaction to volatility shocks.

9.2 Supertrend Indicator

Supertrend combines the average true range (ATR) with price position to define adaptive bands. Define:

$$\mathrm{ATR}_t = \mathrm{EMA}(|H_t - L_t|, N) \tag{25}$$

$$\mathrm{Upper}_t = \mathrm{HL2}_t + M \cdot \mathrm{ATR}_t \tag{26}$$

$$\mathrm{Lower}_t = \mathrm{HL2}_t - M \cdot \mathrm{ATR}_t \tag{27}$$

$$\mathrm{HL2}_t = \frac{H_t + L_t}{2} \tag{28}$$

where H_t, L_t are high/low at t, M multiplier (typically 2-3). Supertrend switches signals on crossovers, adapting to realized volatility.

9.3 Connection to Volatility

Although primarily used for trend identification, both DEMA and Supertrend serve as volatility filters. Their crossover signals often align with volatility regime changes, providing context for

decision modules downstream.

10 Numerical Example: Volatility, Hedging, and Forecasting in SPY, VIXY

To illustrate the interplay between implied volatility estimation, forecasting, and hedging, consider the case of the SPDR S&P 500 ETF (SPY). The market price of SPY options encodes implied volatility (IV) through the mechanisms described in previous sections. To hedge volatility exposure, traders may use ETFs such as VIXY, which track the performance of index futures on volatility.

10.1 Option Pricing and Implied Volatility

Suppose at time t, SPY trades at $S_t = \$450$, the 30-day at-the-money call option trades for $C_{\text{mkt}} = \$8.75$. Assume $r = 0.02$ annualized, and time to expiration $T = 30/252 \approx 0.119$. To find σ_{imp}, we solve:

$$C_{\text{mkt}} = C_{\text{BS}}(S_t, K, T, r, \sigma_{\text{imp}}) \tag{29}$$

using numerical inversion. Suppose $\sigma_{\text{imp}} = 0.19$ (or 19%).

10.2 Forecasting Volatility with LSTM

An LSTM network is trained on past SPY returns and volatility estimates, using rolling windows of 60 trading days to forecast the 30-day realized volatility. Suppose the network predicts $\hat{\sigma}_{\text{future}} = 0.22$ with uncertainty quantified (e.g., MC Dropout) as 0.04.

10.3 Hedging Volatility Exposure Using VIXY

The market expectation for volatility can be hedged using VIX derivatives. For example, supposing the trader sells call options on SPY with IV at 19%, but expects realized volatility at 22%, he may buy VIXY shares—whose value rises as realized volatility outpaces market expectation.

The effectiveness of the hedge depends on the correlation structure between SPY returns and VIX futures. The hedge ratio is optimally set by minimizing variance of total portfolio, solution to:

$$\min_{\beta} \operatorname{Var}\left[P_{SPY} + \beta P_{VIXY}\right] \tag{30}$$

where P_{SPY} and P_{VIXY} are profit and loss from SPY and VIXY positions.

10.4 Dynamic Position Sizing and RL Exit

Position sizes are set as functions of ensemble regime detection, forecasted volatility, and technical analysis signals. Suppose regime classifier detects "bullish trend", LSTM volatility is elevated, and DEMA/Supertrend support long exposure. RL agent monitors real-time P&L, regime changes, and volatility estimate, updating the optimal stopping boundary (as per its Q-values).

At each major decision point, the agent compares realized profit, time held, and regime stability, and executes exit if expected reward falls below threshold adjusted for volatility.

11 Spectral Clustering for Implied Volatility Skew Prediction

In this section, the implied volatility skew is modeled as the manifestation of latent regimes on a graph whose vertices represent option contracts and whose edges encode similarity of their volatility characteristics. Spectral clustering is used to extract these regimes from the graph Laplacian, and the resulting clusters are then employed to forecast future skew dynamics, both cross-sectionally (across strikes) and term-structurally (across maturities).

11.1 Graph Construction from Option Features

Let each option contract at time t be represented by a feature vector

$$x_i(t) = \big(m_i(t), \tau_i(t), \sigma_i^{\text{imp}}(t), \Delta_i(t), \text{IVSlope}_i(t), \dots\big) \in \mathbb{R}^d,$$

where $m_i(t)$ denotes moneyness S_t/K_i, $\tau_i(t)$ the time to expiry in years, $\sigma_i^{\text{imp}}(t)$ the observed implied volatility, and $\text{IVSlope}_i(t)$ a local estimate of the skew (e.g. a finite difference approximation $\partial\sigma^{\text{imp}}/\partial K$). The option universe at time t forms a vertex set

$$V_t = \{1, 2, \dots, N_t\}, \qquad X_t = \{x_i(t)\}_{i=1}^{N_t}.$$

A weighted, undirected graph $G_t = (V_t, E_t, W_t)$ is built by defining pairwise similarities via a kernel

$$w_{ij}(t) = \exp\left(-\frac{\|x_i(t) - x_j(t)\|_2^2}{2\gamma^2}\right) \mathbf{1}\{\|x_i(t) - x_j(t)\|_2 \le \varepsilon\},$$

where $\gamma > 0$ controls the scale of the Gaussian kernel and ε is a sparsification radius. The adjacency matrix $W_t = (w_{ij}(t))_{1\le i,j\le N_t}$ encodes similarity of contracts in terms of their joint moneyness, maturity and volatility characteristics.

11.2 Spectral Embedding and Normalized Graph Laplacian

Define the degree matrix $D_t = \mathrm{diag}(d_1(t), \ldots, d_{N_t}(t))$ with

$$d_i(t) = \sum_{j=1}^{N_t} w_{ij}(t),$$

and consider the symmetric normalized graph Laplacian

$$L_t = I - D_t^{-1/2} W_t D_t^{-1/2}.$$

The matrix L_t is positive semidefinite and admits an eigendecomposition

$$L_t u_k(t) = \lambda_k(t) u_k(t), \qquad 0 = \lambda_1(t) \leq \lambda_2(t) \leq \cdots \leq \lambda_{N_t}(t).$$

For a prescribed number of clusters K, the spectral embedding is constructed from the first K nontrivial eigenvectors:

$$U_t = \big[u_1(t), u_2(t), \ldots, u_K(t)\big] \in \mathbb{R}^{N_t \times K}.$$

The i-th row of U_t, denoted $z_i(t) \in \mathbb{R}^K$, embeds option i into a low-dimensional space where clusters corresponding to coherent volatility regimes become approximately linearly separable.

11.3 Clustering of Volatility Regimes

A standard K-means algorithm is applied to the rows $\{z_i(t)\}_{i=1}^{N_t}$ to obtain a partition

$$\mathcal{C}_t = \{C_1(t), C_2(t), \ldots, C_K(t)\}$$

of the option universe into K clusters. Each cluster $C_k(t)$ can be interpreted as a volatility regime characterized by a typical skew profile and level of implied volatility.

For a given cluster $C_k(t)$, define a representative cluster skew function at time t by

$$\widehat{\sigma}_k(m, \tau; t) = \mathrm{median}\big\{\sigma_i^{\mathrm{imp}}(t) : i \in C_k(t),\, m_i(t) \approx m,\, \tau_i(t) \approx \tau\big\},$$

where the median is taken over options in $C_k(t)$ whose moneyness and maturity fall in a small neighborhood of (m, τ). The collection $\{\widehat{\sigma}_k(\cdot, \cdot; t)\}_{k=1}^K$ yields a set of prototypical implied volatility surfaces conditional on regime.

11.4 Temporal Dynamics and Regime Transition Probabilities

As time evolves, the spectral clustering step is repeated on a rolling window of surfaces, yielding a sequence of partitions $\{\mathcal{C}_t\}_{t \in \mathcal{T}}$. Let $R_t \in \{1, \ldots, K\}$ denote the dominant cluster (regime) at time t, for example defined as the cluster capturing the largest notional vega or open interest.

The empirical regime transition matrix $P \in \mathbb{R}^{K \times K}$ is estimated as

$$P_{ab} = \mathbb{P}(R_{t+1} = b \mid R_t = a) \approx \frac{\#\{t : R_t = a,\, R_{t+1} = b\}}{\#\{t : R_t = a\}},$$

which defines a finite-state Markov chain on the set of volatility regimes. This Markov structure enables probabilistic forecasts of future skew configurations by propagating the current regime forward using P.

11.5 Forecasting Implied Volatility Skew

Given the current regime $R_t = a$, the one-step-ahead risk-neutral forecast of the skew at moneyness m and maturity τ is obtained by averaging the regime-specific prototypical surfaces under the transition probabilities:

$$\mathbb{E}[\sigma^{\text{imp}}(m, \tau; t+1) \mid R_t = a] \approx \sum_{b=1}^{K} P_{ab}\, \widehat{\sigma}_b(m, \tau; t),$$

with higher-horizon forecasts using powers of P, i.e. P^h for horizon h. This provides a nonparametric, data-driven alternative to parametric skew models such as SVI, where the dynamics of skew are captured by transitions between learned spectral clusters rather than by explicit stochastic differential equations for a finite number of shape parameters.

11.6 Visualization of Spectral Embeddings and Clustered Surfaces

For interpretability, the spectral embedding can be visualized in three dimensions by plotting the components of $z_i(t)$ for each option and coloring points by their assigned cluster. Figure 2 shows the resulting 3D embedding, which reveals how options group into distinct volatility regimes in the reduced space.

Figures 3 and 1 display the corresponding cluster-conditional and global implied volatility surfaces derived from these embeddings.

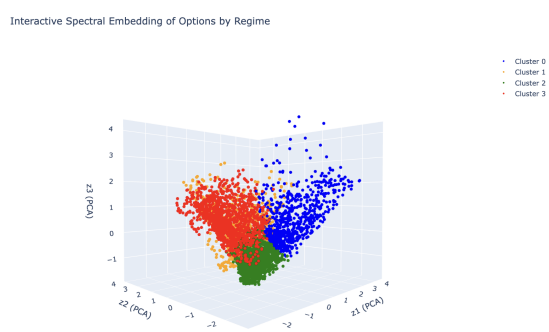
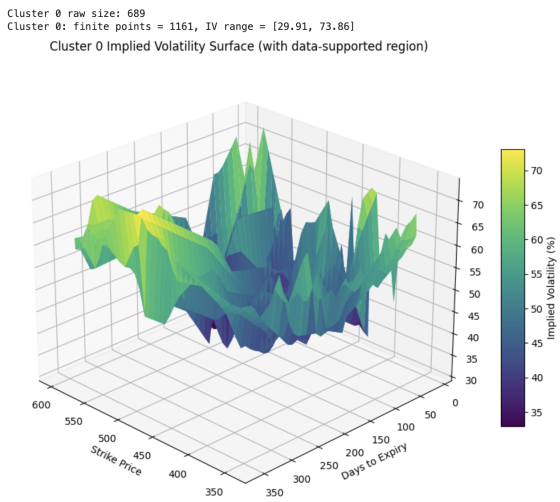
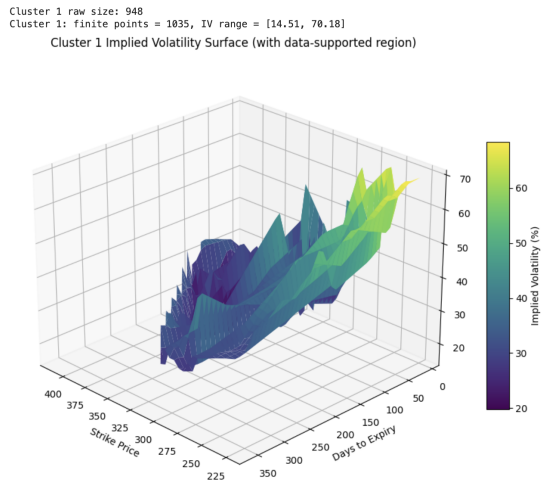


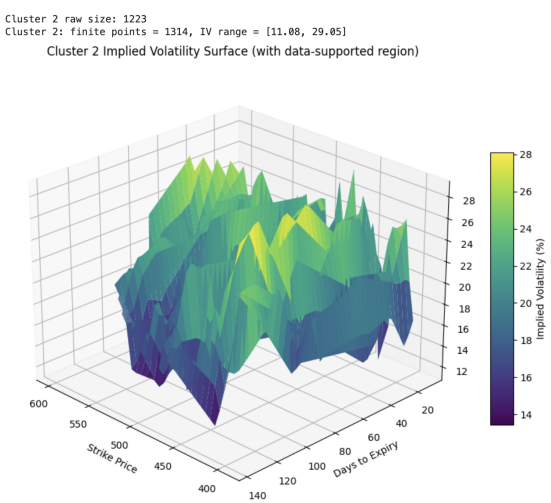
Figure 2: Three-dimensional spectral embedding of options, with points colored by cluster assignment. The geometry highlights regime separation in the latent space. For full code and data, see this Kaggle notebook.



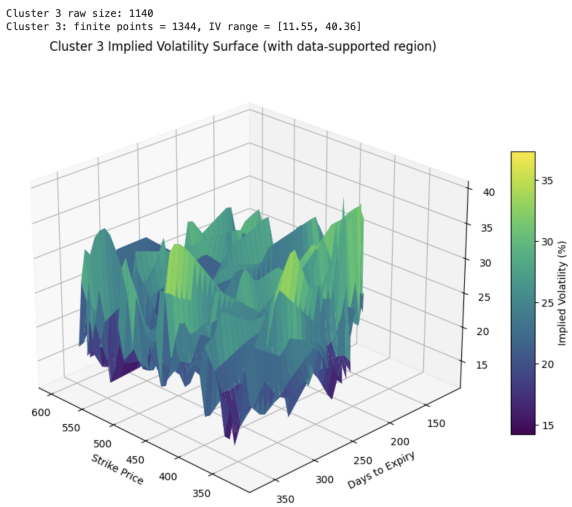
Cluster 0 surface



Cluster 1 surface



Cluster 2 surface



Cluster 3 surface

Figure 3: Cluster-conditional implied volatility surfaces by spectral regime (top left to bottom right: Clusters 0–3).

12 Synthesis: End-to-End Regime-Aware Volatility Trading Algorithm

This section synthesizes the preceding components into a single, fully specified algorithm for regime-aware volatility trading. The architecture integrates spectral clustering of implied volatility surfaces, LSTM-based volatility forecasting, and deep reinforcement learning for execution, with an explicit risk and calibration layer. Information flow, decision rules, and training and evaluation procedures are detailed so that the framework can be implemented without further interpretation.

12.1 Information Flow and System State

Time is indexed in trading days $t = 1, \dots, T$. At each t, the system ingests market data, updates regimes and forecasts, constructs a state vector s_t, and passes this state to a policy that chooses an action a_t.

12.1.1 Market data and features.

For each option contract i on SPY at time t, the feature vector

$$x_i(t) = \big(m_i(t), \tau_i(t), \sigma_i^{\text{imp}}(t), \Delta_i(t), \text{IVSlope}_i(t), \dots\big) \in \mathbb{R}^d$$

is computed as in Section 11. Here $m_i(t)$ is moneyness, $\tau_i(t)$ time to maturity, $\sigma_i^{\text{imp}}(t)$ implied volatility, $\Delta_i(t)$ option delta, and $\text{IVSlope}_i(t)$ a local estimate of skew.

12.1.2 Spectral regimes and regime surfaces.

Given the option universe $X_t = \{x_i(t)\}$, a similarity graph G_t is constructed with weights

$$w_{ij}(t) = \exp\left(-\frac{\|x_i(t) - x_j(t)\|_2^2}{2\gamma^2}\right) \mathbf{1}\{\|x_i(t) - x_j(t)\|_2 \le \varepsilon\},$$

from which the normalized Laplacian L_t and its first K nontrivial eigenvectors yield the spectral embedding $z_i(t)$. Clustering $\{z_i(t)\}$ with K-means produces partitions $\mathcal{C}_t = \{C_k(t)\}_{k=1}^K$ and a dominant regime label $R_t \in \{1, \dots, K\}$ for the market, for example defined as the cluster with largest notional vega or open interest.

For each cluster k, a regime-conditional implied volatility surface is estimated by

$$\widehat{\sigma}_k(m, \tau; t) = \text{median}\big\{\sigma_i^{\text{imp}}(t) : i \in C_k(t),\ m_i(t) \approx m,\ \tau_i(t) \approx \tau\big\}.$$

Over a historical training window $\mathcal{T}_{\text{train}}$, the regime sequence $\{R_t\}$ defines the empirical transition matrix

$$P_{ab} = \frac{\#\{t \in \mathcal{T}_{\text{train}} : R_t = a,\ R_{t+1} = b\}}{\#\{t \in \mathcal{T}_{\text{train}} : R_t = a\}}, \qquad a, b \in \{1, \dots, K\},$$

which specifies a finite-state Markov chain on volatility regimes.

12.1.3 LSTM-based realized volatility forecast.

Let σ_t^{real} denote realized volatility over a fixed horizon (e.g. 30 days). An LSTM f_θ is trained on sequences

$$\big(r_{t-n}, \ldots, r_t,\ \sigma_{t-n}^{\text{real}}, \ldots, \sigma_t^{\text{real}},\ R_{t-n}, \ldots, R_t\big)$$

to predict $\hat{\sigma}_{t+\Delta}^{\text{real}} = f_\theta(\cdot)$, yielding a point forecast $\hat{\sigma}_{\text{LSTM}}(t)$ and, if desired, an uncertainty estimate $\hat{\sigma}_{\text{LSTM,std}}(t)$ via techniques such as Monte Carlo dropout.

12.1.4 Regime-mixture implied surface forecast.

Given current regime $R_t = a$, the one-step-ahead implied surface is forecast via the Markov mixture

$$\hat{\sigma}^{\text{imp}}(m, \tau; t+1 \mid R_t = a) = \sum_{b=1}^{K} P_{ab}\, \widehat{\sigma}_b(m, \tau; t),$$

from which low-dimensional summaries are extracted, such as an at-the-money level ℓ_t and skew skew_t at reference (m^*, τ^*).

12.1.5 Technical indicators and portfolio greeks.

Technical features $I_t \in \mathbb{R}^q$ are computed from DEMA, Supertrend, and other volatility-sensitive indicators, while current portfolio exposures are summarized by a vector of greeks $\Gamma_t = (\Delta_t, \text{Vega}_t, \Gamma_t, \ldots)$.

12.1.6 State vector.

All information available to the decision module is aggregated into a state

$$s_t = \big(R_t,\ \ell_t,\ \text{skew}_t,\ \hat{\sigma}_{\text{LSTM}}(t),\ \hat{\sigma}_{\text{LSTM,std}}(t),\ I_t,\ \Gamma_t\big),$$

which serves as the input to the policy and value function.

12.2 Action Space, Policy, and Reward

12.2.1 Action space.

Trading is defined over a fixed library of option templates $\{\mathcal{T}_j\}_{j=1}^{J}$ (e.g. ATM straddles, 25-delta risk reversals, calendar spreads). The action space is

$$\mathcal{A} = \{\text{Flat}\} \cup \big\{\pm q_k\, \mathcal{T}_j : j = 1, \ldots, J,\ k = 1, \ldots, K_q\big\},$$

where q_k are standardized position sizes (e.g. small, medium, large), and the sign indicates long or short volatility exposure. Each non-flat action thus corresponds to opening, increasing, or decreasing a position in a specified structure at a standardized notional.

12.2.2 Policy and value function.

A parameterized policy $\pi_\phi(a \mid s)$ maps states to action probabilities. In an actor–critic implementation, a value function $V_\psi(s)$ approximates expected discounted return, while the actor updates ϕ along the policy gradient. In a value-based implementation such as DQN, $Q_\phi(s, a)$ approximates the state–action value and the behavior policy is ε-greedy:

$$a_t = \arg\max_{a \in \mathcal{A}} Q_\phi(s_t, a) \quad \text{with probability } 1 - \varepsilon_t,$$

and a random action otherwise.

12.2.3 Reward function.

Let Π_t denote portfolio value, $\Delta\Pi_{t+1} = \Pi_{t+1} - \Pi_t$ the one-step P&L, and Costs_{t+1} total transaction costs incurred over $[t, t+1]$. The reward is defined as

$$r_{t+1} = \Delta\Pi_{t+1} - \lambda_{\text{risk}} \operatorname{Var}_t(\Pi_{t+1}) - \lambda_{\text{tc}} \, \text{Costs}_{t+1},$$

where $\operatorname{Var}_t(\Pi_{t+1})$ is approximated from local greeks and underlying/implied-volatility variance. Optionally, a volatility-spread term can be included to explicitly reward profitable exploitation of IV–RV deviations,

$$+ \lambda_{\text{spread}} \big(\hat{\sigma}_{\text{LSTM}}(t) - \sigma_{\text{imp,traded}}(t)\big) \cdot \text{Vega}_{\text{signed},t},$$

where $\sigma_{\text{imp,traded}}$ is the implied volatility of the traded structure and $\text{Vega}_{\text{signed},t}$ its signed vega.

12.3 Risk Management and Tightness Calibration

Risk control is embedded both as hard constraints on admissible actions and as soft penalties in the objective.

12.3.1 Position and exposure limits.

At each time t, candidate actions are filtered by

$$|\Delta_t| \le \Delta_{\max}, \qquad |\text{Vega}_t| \le V_{\max}, \qquad \text{GrossNotional}_t \le L_{\max} \cdot \text{Equity}_t,$$

with any action that would violate a constraint projected to the nearest admissible action or replaced by Flat.

12.3.2 Regime-contingent safety overrides.

A subset $\mathcal{R}_{\text{stress}} \subset \{1, \ldots, K\}$ of crisis regimes is identified based on historical averages of implied volatility and skew. If $R_t \in \mathcal{R}_{\text{stress}}$, the action space is restricted to a conservative subset $\mathcal{A}_{\text{safe}} \subset \mathcal{A}$ (e.g. hedging or flat positions), and maximum holding periods may be shortened.

12.3.3 Drawdown and stop rules.

Let E_t denote equity and $E_t^* = \max_{u \le t} E_u$ the running peak. If $E_t < (1 - D_{\max})E_t^*$, trading is halted and the policy is forced to Flat for a cooldown horizon H_{cool}, during which only observation and parameter updates occur. Analogous stops can be set on daily P&L or realized volatility spikes.

12.3.4 Tightness via execution thresholds.

The system executes trades only when volatility discrepancies exceed calibrated thresholds. A canonical rule is

$$\left|\hat{\sigma}_{\text{LSTM}}(t) - \sigma_{\text{imp,traded}}(t)\right| > \delta,$$

for some $\delta > 0$ chosen by out-of-sample optimization, reflecting the minimum mispricing required to dominate transaction costs and model uncertainty. When this condition is not met, the effective action becomes Flat, regardless of the policy suggestion. More generally, tightness parameters are collected in a vector θ (execution thresholds, regime filters, position caps), and tuned via

$$\min_{\theta} \mathbb{E}\Big[L\big(\text{decisions}(\theta),\ \text{market reality}\big) + \lambda\|\theta - \theta_0\|_2^2\Big],$$

where L encodes a risk-adjusted performance loss and θ_0 encodes baseline settings.

12.4 Training Procedure

Training proceeds in three stages: unsupervised regime learning, supervised volatility forecasting, and reinforcement learning for control.

12.4.1 Stage 1: spectral regimes and transition matrix.

Using historical options data on a training period $\mathcal{T}_{\text{train}}$, graphs, Laplacians, embeddings, and clusters are constructed at each t, yielding regime labels R_t and regime-conditional surfaces $\widehat{\sigma}_k(m, \tau; t)$. The transition matrix P is estimated empirically from $\{R_t\}$ as above.

12.4.2 Stage 2: LSTM volatility model.

Realized volatility σ_t^{real} is computed over rolling windows, and the LSTM is trained to minimize a forecasting loss such as mean squared error,

$$\min_{\theta} \sum_{t \in \mathcal{T}_{\text{train}}} \left(\sigma_{t+\Delta}^{\text{real}} - f_\theta(\text{inputs}_t)\right)^2,$$

where inputs_t comprises returns, past realized volatility, and regime labels. Hyperparameters (layers, hidden units, regularization) are selected via cross-validation and, if desired, compared against GARCH or HAR baselines.

12.4.3 Stage 3: reinforcement learning in historical environment.

A historical environment is constructed by replaying market data over $\mathcal{T}_{\text{train}}$ (or a sub-interval reserved for RL), with spectral and LSTM modules treated as fixed. At each step:

1. Compute state s_t from current regimes, forecasts, indicators, and portfolio greeks.
2. Select action a_t using an exploratory policy (e.g. ε-greedy or stochastic actor).
3. Apply risk filters and tightness thresholds, adjust a_t to an admissible action.
4. Compute next state s_{t+1} and reward r_{t+1} from realized P&L and penalties.
5. Store transition $(s_t, a_t, r_{t+1}, s_{t+1})$ in a replay buffer.

Parameters ϕ (and ψ, for actor–critic) are updated by sampling minibatches from the buffer and performing stochastic gradient updates to approximate the optimal policy π^* maximizing expected discounted reward. Exploration rate ε_t is scheduled to decay while preserving sufficient randomization to explore different regimes.

12.5 Evaluation and Outlook

12.5.1 Backtesting protocol.

On an out-of-sample test set $\mathcal{T}_{\text{test}}$, the full system operates without parameter updates other than those explicitly allowed by the design (e.g. periodic re-estimation of P and LSTM fine-tuning). Performance metrics include annualized return, Sharpe ratio, maximum drawdown, turnover, and realized vs implied volatility spread exploitation, evaluated both overall and conditioned on regime.

12.5.2 Baselines and ablations.

Benchmark strategies include: (i) a parametric volatility-surface model (e.g. SVI+GARCH) with simple IV–RV spread trades; (ii) an LSTM-only volatility strategy without spectral regimes; and

(iii) a deep RL price-based trader without access to implied volatility surfaces. Ablation studies remove or replace individual components (spectral clustering, LSTM, RL, risk layer) to quantify their marginal contribution to risk-adjusted performance and robustness across regimes.

12.5.3 Perspective.

The resulting architecture offers a unified, volatility-centric view of trading in which implied volatility surfaces are not merely calibration objects but evolving geometric entities that define regimes, drive forecasts, and constrain decisions. Spectral clustering provides a nonparametric representation of volatility structure, LSTMs supply regime-aware forecasts of future variability, and reinforcement learning transforms these forecasts into adaptive trading policies under explicit risk and tightness constraints. In this sense, the framework constitutes a mathematically grounded algorithmic lens on markets where volatility is both the central signal and the principal object of control.